Networks and Complexity

Exercise Sheet 23: No current without heat

This is an exercise sheet from the forthcoming book Networks and Complexity. Find more exercises and solutions at https://github.com/NC-Book/NCB

Ex 23.1: SIS on a three node chain [Lvl. 1]

In the chapter we introduced the interacting particle notation for the SIS model master equation. Use your intuition to formulate the Jacobian of the master equation for the SIS model on a chain of three nodes, (1)-(2)-(3).

Ex 23.2: Counting spanning trees [Lvl. 2]

Use the matrix-tree theorem to count the number of spanning trees in a fully connected network of 4 nodes. Can you also count them in a fully connected network of n nodes?

Ex 23.3: Chain and cycle [Lvl. 2]

Let's study another biased random walk.

a) Consider a walker on a 3-chain, (1)-(2)-(3). When in a node $i < 3$, the walker moves to node $i + 1$ at rate 2. When in a node $i > 1$ it moves to node $i - 1$ at rate 1. Determine the probabilities of finding the walker in node i in the stationary state of the system.

b) Now close the chain to form a triangle. In addition to the transitions from part (a), a walker that is in node 3 moves directly to node 1 at rate 2 and a walker that is in node 1 moves directly to node 3 at rate 1. Again find the stationary state.

c) Show that the detailed balance condition is met in the stationary state in the network from (a) but not in the network from (b).

Ex 23.4: Customers [Lvl. 3]

In a shop customers arrive at an average rate of a customers per minute. Customers in the shop leave at a rate of b customers per customer per minute. Use detailed balance to compute x_n the probability that there are n customers in the shop, in the stationary state.

Ex 23.5: Paper supply [Lvl. 3]

In an office the copier paper is stored in a cabinet. At an average rate of once a week an employee takes paper out of the cabinet and loads it into the copier. There is a 50% chance that the employee tries to load only one packet of paper, and a 50% chance that the employee tries to load two packets of paper. If the cabinet is empty, the office manager immediately orders 4 new packets of paper which arrive on average within 2 weeks.

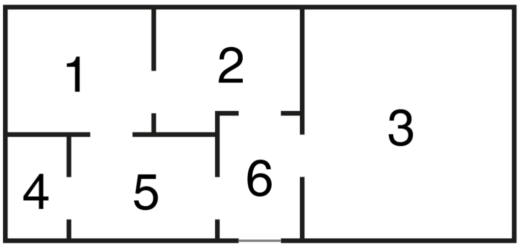
a) Consider this system as a network of states, numbered 0 to 4. The system is in state i if there are i packets of paper in the cabinet. Write a system of differential equations for the variables $x_0, \ldots, x_4$, where x_i is the probability that the system is in state i.

b) Write the differential equation system in the form $\dot{\boldsymbol{x}} = -\mathbf{L}\boldsymbol{x}$ where $\boldsymbol{x} = (x_0, \ldots, x_4)^{\mathrm{T}}$.

c) Use Kirchhoff's theorem to compute the probability that there isn't any paper in the cabinet. (i.e. compute x_0 in the steady state).

Ex 23.6: Cat model [Lvl. 3]

A cat roams freely in the flat shown below. At rate 1 the cat becomes bored of the room she is currently in, then she picks one of the doors of the current room at random and walks through it. Determine the probability x_n that the cat is in room n in the stationary state.



Ex 23.7: Well-mixed SIS [Lvl. 3]

Consider SIS dynamics on a network, with one additional process: If there are no infected then a random node is infected at rate a from an external source.

a) Consider the SIS epidemic on the linked-pair. Use the symmetry to argue that in the case where one agent is infected, it does not matter who is infected, and hence formulate a master equation for the variables x_0, x_1, x_2, that are the probabilities of having 0, 1, or 2 infected.

b) Find the steady state of your model on the linked pair, by exploiting the detailed balance property.

c) Now consider the same epidemic process on a fully connected network of three nodes. Use Kirchhoff's method to compute the probability of finding three, two, one or none infected agents in the stationary state.

d) Finally, consider the epidemic on a fully-connected network of N nodes and determine the probability that n agents are infected in the stationary state.

Ex 23.8: Watching a kettle boil [Lvl. 3]

Let's explore if observing the system can actually change its time evolution.

a) Consider a diffusion process where a walker moves randomly between the nodes of the linked-pair network so that jumps occur on average at rate 1. The walker starts at time zero in node 1. Compute the probability $x_1(2)$ that the walker is in node 1 at time 2.

b) Now assume that we check on the walker at time 1. With probability $x_1(1)$ we find it in node 1, with probability $x_2(1) = 1 - x_1(1)$ we find it in node 2. Of course, the measurement resets the probabilities. For example, if we see the walker in node 1 at time 1, we can be sure that it is there and hence $x_1(1)$ becomes 1. Does knowing where the walker is at time 1 change the chance that it will be in node 1 at time 2?

Ex 23.9: Fixation probability [Lvl. 3]

In a flowerbed there is space for five plants. Plants die at a rate of 3 per plant per time. When this happens, one of the other plants has space to reproduce, so the space freed by the dying plant is filled (immediately) by an offspring of one of the other plants, chosen at random. But one plant has a rare mutation that makes it more resilient such that this plant only dies at rate 2. All of the mutant plant's offspring have the same mutation, whereas neither of the offspring of other plants has it. Compute the probability that mutant plants take over the whole flowerbed.

Ex 23.10: Two walkers [Lvl. 4]

Consider two distinguishable non-interacting random walkers on a network with Laplacian matrix $\mathbf{L}$, such that the master equation for each walker individually is $\dot{\boldsymbol{x}} = -\mathbf{L}\boldsymbol{x}$ where x_i is the probability that the walker is in node i. Then define y_{ij} as the probability that the first walker is in node i and the second walker is in node j.

a) Consider the system on the linked pair graph. Write differential equations for y_{ij}. Use Kronecker products to compactly write the Jacobian of the system and use it to find the general solution of the system.
b) For the network from (a), define z_i as the expectation value of walkers in node i. Translate your solution for $y_{ij}(t)$ into a solution for $z_i(t)$. Can you also identify the differential equation that $\boldsymbol{z} = (z_1, z_2)^{\mathrm{T}}$ follows?
c) Repeat your approach from parts (a) and (b) but instead of a linked pair, focus on two walkers on a general network topology, described by an unspecified Laplacian $\mathbf{L}$.

Ex 23.11: Lyapunov property [Lvl. 4]

Consider a dynamical system of the form $\dot{\boldsymbol{x}} = -\mathbf{L}\boldsymbol{x}$, where $\boldsymbol{x} = (p_1, p_2, \ldots)^{\mathrm{T}}$, and $\mathbf{L}$ is a Laplacian matrix. Let $\boldsymbol{x}^*$ be a stationary state, and $S_{\mathrm{KL}} = \sum p_k \log p_k/p_k^*$. Show the following:
a) $S_{\mathrm{KL}} = 0$ if $\boldsymbol{x} = \boldsymbol{x}^*$
b) $\dot{S}_{\mathrm{KL}} = 0$ if $\boldsymbol{x} = \boldsymbol{x}^*$
c) $S_{\mathrm{KL}} > 0$ if $\boldsymbol{x} \neq \boldsymbol{x}^*$
d) $\dot{S}_{\mathrm{KL}} < 0$ if $\boldsymbol{x} \neq \boldsymbol{x}^*$ (Hard. Check the solution if you get stuck.)

Ex 23.12: Kirchhoff proof [Lvl. 5]

Find a forward-facing derivation of Kirchhoff's theorem and the matrix-tree theorem, i.e. a line of reasoning that leads to the result, rather than postulating the result and then proving that it is true.